

A SYSTEMATIC REVIEW OF ELECTRONIC TOLL COLLECTION SYSTEMS

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ABSTRACT

The Electronic toll collection (ETC) is typically defined as wireless system to automatically collect the usage fees or tolls from vehicles. The ETC is a new system that provides a smooth experience to the passengers by giving them a traffic free highway. The most notable advantage of this technology is that it eliminates congestions near the toll booths. The ETC ensures smooth public transports, uninterrupted traffic flows, reduced congestion, better environment, revenue generation and road safety as well. This paper aims to synthesize the effective technologies of the ETC systems and its necessity in highway traffic management system in a developing country like Bangladesh. There are many challenges in introducing ETC in Bangladesh. As this is a new technology, it would take time to overcome the difficulties. Efforts from all corners can move the ETC forward and make excellent use of the modern and digital technology in the payment of toll in line with the viewpoint of the digital Bangladesh. The ETC implementation is costly but in the long run it will give a sustainable highway traffic management. However, Dedicated Short Range Communication (DSRC), Radio Frequency Identification (RFID) based ETC which are relatively easy to implement and also cheaper than others can be implemented in Bangladesh in a short term.

Keywords: Electronic Toll Collection (ETC), Radio Frequency Identification (RFID), Dedicated Short Range Communication (DSRC), Digital Bangladesh.

INTRODUCTION

The Electronic Toll Collection (ETC) is a modern technology that allows vehicles to pass the toll booths without stopping the vehicle for payment. The main advantage of the ETC is to ensure uninterrupted travel near the toll plaza. Additionally, the maintenance cost of the toll booth is also reduced as there is no necessity for staff in the booth. It is a fully automated technology that reduces the transaction time. Reduced traffic congestion near the toll plaza is one of the main core motivations of implementing the electronic toll system. The traffic congestion on the highway is a common scenario in developing countries. Therefore, implementing ETC is a demand based need to reduce congestion. This digital technology saves a huge amount of time as the vehicles leaving the booth do not need to halt to give a toll, and uses Short Message Service (SMS) instead of receipts (Khan F. H., 2019)

The ETC is not a new concern. Back in 2004/05 Santiago, Chile implemented the world's first 100% full speed electronic tolling with transponder. The United Arab Emirates implemented a similar road toll collection in 2007 in Dubai. However, the ETC is not well standard as like as developed countries. The implementation of ETC in developing countries is more difficult because of it's complex technologies. The developing country also faces additional challenge (due to corruption) while collecting the tolls from manual toll system.

The transport sector plays an important role in Bangladesh's economy. About 12% of GDP and 20% of the annual development budget has been spent in the transport sector. At the current growth, the number of vehicles in the country is expected to be double in the next ten years (Hoque et al, 2007). This amount of vehicle creates a heavy pressure on the highways and expressways. The authority is trying to solve this crisis by taking mega projects. The Intelligent Transportation System (ITS) has already implemented in Bangladesh as part of a pilot based project (Khan, Sakib et al, 2014).

This paper aims to synthesize the effective technologies of ETC systems, its necessity in highway traffic management system in a developing country like Bangladesh.

COUNTRIES ADOPTING ETC

Several countries have already introduced the ETC technology in the transportation systems. Table 1 includes the ETC technologies and related factors with corresponding countries.

Table 1: The ETC Systems in Different Countries

COUNTRY	TECHNOLOGY	FACTORS AND RELATED TECHNOLOGIES
Singapore	Singapore will implement Satellite-based Electronic Road Pricing System (ERP) from the first half of 2021.	GNSS, RFID, Cash card, Smartcard, Land Transport Authority (LTA)
Japan	Japanese car navigation system, VICS (Vehicle Information and Communication System)	DSRC, ITS, ETC, Ministry of Land, Infrastructure and Transport of Japan
Germany	The German Toll Collection GPS-GSM system	GSM, OBU, Automatic Payment Method
Hong Kong	The Automated Toll Collection was first introduced in Hong Kong in 1993 with DSRC Technologies.	OBU, DSRC (Dedicated short range communication), ITS
Stockholm	Congestion pricing system implemented in Stockholm, Sweden.	Stockholm Congestion Tax, Traffic congestion
China	China is working on 5G-enable C-V2X networks.	AV-CV technology, ETC device
America	Advanced technology like End-To-End Service Crowded Toll and AuditToll DBE.	RFID, ETC, DSRC, ANPR

ETC RELATED TECHNOLOGIES

The ETC has already implemented in many countries as an efficient transportation system. Advanced technologies have already been developed in ETC system. Table 2 includes a comprehensive literature review of the ETC technologies with corresponding methods and its impacts.

Table 2: The ETC Technologies with Corresponding Methods and Impacts

NAME	METHOD	IMPACT	AUTHORS (YEAR)
Automatic Toll Collection (ATC)	The use of the Automated Coin Machine is the basis for automatic toll collection (ACM). Both coins and tokens issued by the operating agency recognize these. The use of automated coin or token collection instead of manual collection decreases transaction and processing time, as well as operational costs.	Day by day ATC replaces Manual Toll Collection (MTC). The reason is that ATC does not need the drivers to slow down the vehicles in toll booths for cash.	Sharpe et al. (1996) Salunke et al. (2013)
Automatic Number Plate Recognition (ANPR)	The technology of optical character recognition on images to read vehicle registration plates to produce vehicle position data is the automatic number plate recognition (ANPR). ANPR is used for law enforcement purposes by police forces around the world, including verification of the registered or the licensed vehicle.	This system identifies unregistered and stolen vehicles also disqualified or suspended drivers. It can be also used for speed limit enforcement. It makes tracking cars easier.	Goyal et al. (2016) Ajanthan et al (2013)
Barcode-based ETC	A barcoded sticker is added to the vehicle and read as it passes through the toll plaza by a laser scanner, illustrating the function of a standard electronic tolling DSRC device.	It is both the simplest and the oldest technology.	Sanchit et al. (2014)
RFID based ETC	The easiest technology for ETC is Radio-frequency identification (RFID) which has an IVU (In-vehicle unit) installed on the front windshield of the vehicle. It contains a cash card for payment the toll. At toll booth, this is read by the RFID frequency reader. It can be prepaid or postpaid.	Comparatively it is cheapest than any other technologies for ETC.	Chattoraj et al. (2017)
DSRC based ETC	Dedicated Short Range Communication (DSRC) is one or two way short to medium range wireless communication uses 5.9 GHz band. It was used in ETC in Japan and Europe in 2003.	DSRC can warn the blind spots, traffic data collection, collision warning, incoming vehicle warning etc.	Anowar et al. (2016)
GNSS based tolling	Global Navigation Satellite System (GNSS) is based on GPS and GMS. It is used for a wide range of application. The principle of tolling is user pays method.	Toll revenue in Germany 2.86 billion euros in 2005.	Numrich et al. (2012)
Cell Phone Tolling	Cell phone tolling is a method that it has potential for mileage-tolling. At first, a chip similar to a cell phone chip would be installed in a vehicle, and frequent communication between cellular towers and the chip would determine how far the car has moved and would assess a toll.	It will be a great uses of 2G mobile phones. This technology is not much expensive.	Vittorio et al. (2013)

FEASIBILITY OF ETC TECHNOLOGIES IN BANGLADESH

Bangladesh is a developing country with remarkable progress in reducing poverty and creating opportunities for its citizens. It is among the ten fastest growing countries in the world and has made commendable progress on human development. The Bangladesh economy is the 32nd largest in the world in terms of purchasing power parity and it is considered to be within the next eleven emerging market economies in the world. The transportation system of Bangladesh is still inefficient due to its heterogeneous network. However, Bangladesh has been witnessing significant technological development since the last decade. Many Asian cities have undergone rising urbanization and private car ownership in recent years, along with other parts of the world, causing an increased number of car accidents, suffering traffic congestion and environmental damage (Anowar et al, 2016). Although Bangladesh needs more time to adapt to the high technological system with the current setting, they can start using ETC gradually (Yeasmin, 2019).

The infrastructure of Bangladesh is the main barrier to implementing ETC Bangladesh. ETC will help for making a database of transport that will be helpful for a better management of traffic control system. As an automated system, there is an option for cutting away. Moreover, ETC will help to reduce the CO² level near the toll plaza and fuel consumption of vehicles (Weng et al., 2015). The Weight-in-motion (WIM) in the ETC system can increase the lifetime of roads (Rahman et al., 2016). Overcoming all obstacles, Bangladesh has already been started to implement ETC in the short range. According to the infrastructure of Bangladesh, RFID based ETC implementation will be more effective as it needs only 2\$ for passive RFID tags (Rahman et al., 2016). A recent survey shows that the 89% of people of Bangladesh believes that ETC would ease traffic jam on highway (Fahmida, 2020). The automated toll collection will be introduced in the Padma Bridge. Moreover, the Dhaka-Mawa-Bhanga expressway is designed with all the facilities of a modern highway (bdnews24.com). The authority should take necessary steps for implementing ETC through the country gradually as the government thinking of collecting tolls on all national highways.

CONCLUSION

In this paper, we have summarised different Electronic Toll Collection (ETC) technologies and its impacts on transport sectors in developing countries like Bangladesh. This paper initially includes the identified relevant technologies of ETC that used in many developed countries. As a developing country Bangladesh has lack of infrastructure in transportation sector. But the associated authority is trying to implement ITS in Bangladesh. The RFID-based ETC which are relatively easy to implement and also cheaper to be implemented in Bangladesh in a short term. But in the long run, implementation of Satellite-based Electronic Road Pricing System (ERP), cell phone tolling etc. can be the best solution both for the authority and the consumers. The number of vehicles will increase day by day. It is high time for thinking about socioeconomic development of Bangladesh for the next ten years and taking necessary steps for implementation of the ETC system.

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